



Mock Class Test 2026

MODULE CODE: COMP202
MODULE TITLE: Complexity of Algorithms

TIME ALLOWED: Forty minutes

	Additional Information
Calculators Allowed?	No
Student permitted materials	Writing materials
Provided materials	Question paper 1x MCQ answer sheet

Instructions for examination:

- This test contributes to 15% of the overall module mark.
- **No toilet breaks are permitted. Exit from the test room for any reason will be treated as final submission of your test, and you will not be allowed to continue.**
- Prepare your answers in the question paper, selecting one option for each question
- When ready, transfer your answers to the MCQ answer sheet provided, shading in the appropriate box for each question

At the end of the examination:

- **Leave the MCQ answer sheet in the examination room**

1. Which of the following is true. **Note:** $\log n = \log(n) = \log_2(n)$.

- ☐ A. $2^n = O(n^{100})$
- ☐ B. $2^n = \Theta(n^{100})$
- ☒ C. $2^n = \Omega(n^{100})$
- ☐ D. $n = O(\log n)$
- ☐ E. $\sqrt{n} = \Omega(2^n)$

2. The function $T(n)$ is assumed to have the initial value

$$T(n) = d, \text{ for any } n \leq c, \text{ where } c, d \geq 1 \text{ are some fixed integer constants.}$$

Note: $\log n = \log(n) = \log_2(n)$.

What is the correct solution to the following recurrence relation that holds for all $n > c$:

$$T(n) = 4T(n/2) + 7n^{8/5}.$$

- ☐ A. $T(n)$ is $\Theta(n^4)$
- ☐ B. $T(n)$ is $\Theta(n^{8/5})$
- ☒ C. $T(n)$ is $\Theta(n^2)$
- ☐ D. $T(n)$ is $\Theta(n^2 \log n)$
- ☐ E. $T(n)$ is $\Theta(n)$

3. The function $T(n)$ is assumed to have the initial value

$$T(n) = d, \text{ for any } n \leq c, \text{ where } c, d \geq 1 \text{ are some fixed integer constants.}$$

Note: $\log n = \log(n) = \log_2(n)$.

What is the correct solution to the following recurrence relation that holds for all $n > c$:

$$T(n) = 9T(n/3) + 10n^{2.6}.$$

- ☐ A. $T(n)$ is $\Theta(n^2)$
- ☒ B. $T(n)$ is $\Theta(n^{2.6})$
- ☐ C. $T(n)$ is $\Theta(n^3)$
- ☐ D. $T(n)$ is $\Theta(n^9)$
- ☐ E. $T(n)$ is $\Theta(n^{10})$

4. The function $T(n)$ is assumed to have the initial value

$T(n) = d$, for any $n \leq c$, where $c, d \geq 1$ are some fixed integer constants.

Note: $\log n = \log(n) = \log_2(n)$.

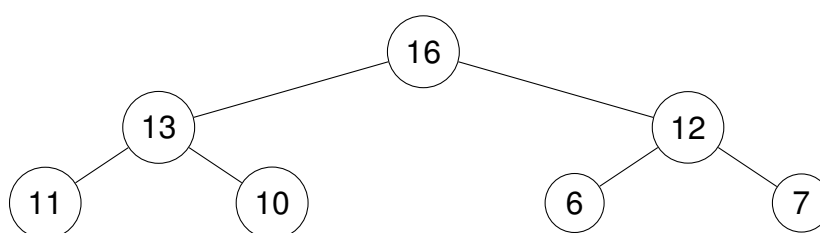
What is the correct solution to the following recurrence relation that holds for all $n > c$:

$$T(n) = 2 T(n/2) + n.$$

- ☐ A. $T(n)$ is $\Theta(n \log^2 n)$
 - ☐ B. $T(n)$ is $\Theta(n^2 \log n)$
 - ☐ C. $T(n)$ is $\Theta(n)$
 - ☒ D. $T(n)$ is $\Theta(n \log n)$
 - ☐ E. $T(n)$ is $\Theta(n^2)$
5. Below is a description of Strassen's matrix multiplication algorithm with some words omitted. "Strassen's algorithm applies to $n \times n$ matrix multiplication, follows a divide-and-conquer approach, and reduces the problem to **XXX** subproblems of size $n/2 \times n/2$, leading to a time complexity of $O(n^{\log_2 \text{XXX}})$." Which of the following is the correct value of **XXX**:
- ☐ A. **XXX** = 5
 - ☐ B. **XXX** = 6
 - ☒ C. **XXX** = 7
 - ☐ D. **XXX** = 8
 - ☐ E. **XXX** = 9
6. Consider an experiment where you keep rolling a fair 6-sided die (with outcomes 1, 2, 3, 4, 5, 6, each equally likely) until you see the value 6 appear, and then you stop the experiment. What is the expected number of rolls you make?
- ☐ A. 1
 - ☒ B. 6
 - ☐ C. 36
 - ☐ D. 3.5
 - ☐ E. 4
7. A bag contains 100 balls, numbers 1 through 100. One ball is selected uniformly at random. What is the probability that the number n on the selected ball satisfies $7 \leq n \leq 16$?
- ☐ A. 1/20
 - ☒ B. 1/10
 - ☐ C. 1/100
 - ☐ D. 1
 - ☐ E. 2

8. What is the best estimate of the expected running time of randomised QUICKSORT for sorting an array of n elements (when pivot is chosen randomly at each step)?
- ☐ A. $O(n^3)$
 - ☐ B. $O(n \log^2 n)$
 - ☐ C. $O(n)$
 - ☒ D. $O(n \log n)$
 - ☐ E. $O(n^2)$
9. Consider the following max-heap T in Figure 1 rooted at the node containing the value 16.

Figure 1:



After performing one REMOVEMAX operation on this heap and restoring the heap structure, which of the following is correct about the new heap obtained:

- ☒ A. The last element in the array representation of the heap is 6
 - ☐ B. The last element in the array representation of the heap is 7
 - ☐ C. The last element in the array representation of the heap is 10
 - ☐ D. The last element in the array representation of the heap is 11
 - ☐ E. The last element in the array representation of the heap is 12
10. What is the best estimate of the running time of an INSERTITEM operation on a heap containing n elements
- ☐ A. $O(n)$
 - ☐ B. $O(2^n)$
 - ☒ C. $O(\log n)$
 - ☐ D. $O(1)$
 - ☐ E. $O(n^2)$
11. Which of the following is the *best* estimate on the worst-case running time of COUNTING SORT on an input of n numbers, each with an integer value between 0 and k .
- ☐ A. $O(n \log n)$
 - ☐ B. $O(n)$
 - ☐ C. $O(k)$
 - ☐ D. $O(nk)$
 - ☒ E. $O(n + k)$

12. You are given the collection of numbers [132, 315, 827, 469, 507, 193, 657, 274] and you wish to sort them in increasing order. After applying the first step of RADIX SORT (i.e., after applying a stable sorting algorithm on the least significant digit), what is the order of the collection?
- ☒ A. [132, 193, 274, 315, 827, 507, 657, 469]
 - ☐ B. [132, 193, 274, 315, 507, 657, 827, 469]
 - ☐ C. [132, 193, 274, 315, 469, 507, 657, 827]
 - ☐ D. [132, 193, 274, 315, 507, 827, 657, 469]
 - ☐ E. [132, 193, 274, 315, 657, 827, 507, 469]
13. Recall that the Fibonacci numbers are defined as $F(0) = 0$, $F(1) = 1$ and $F(j) = F(j - 1) + F(j - 2)$ for all integers $j \geq 2$. Which of the following gives the *best* estimate of the cost of an algorithm that uses repeated squaring to find the n 'th Fibonacci number via repeated squaring of matrices.
- ☐ A. There exists an algorithm that uses repeated squaring of matrices and computes $F(n)$ using $O(2^n)$ matrix multiplications
 - ☐ B. There exists an algorithm that uses repeated squaring of matrices and computes $F(n)$ using $O(n!)$ matrix multiplications
 - ☐ C. There exists an algorithm that uses repeated squaring of matrices and computes $F(n)$ using $O(n^2)$ matrix multiplications
 - ☐ D. There exists an algorithm that uses repeated squaring of matrices and computes $F(n)$ using $O(n)$ matrix multiplications
 - ☒ E. There exists an algorithm that uses repeated squaring of matrices and computes $F(n)$ using $O(\log n)$ matrix multiplications
14. What is the optimal solution in the following instance of the $\{0, 1\}$ **Knapsack Problem** with four items where the maximum allowed weight is $W_{\max} = 10$. Below, i refers to item number, b_i denotes the benefit of item i and w_i denotes the weight of item i .
- | | | | | |
|-------|----|----|----|----|
| i | 1 | 2 | 3 | 4 |
| b_i | 41 | 14 | 17 | 30 |
| w_i | 6 | 2 | 3 | 4 |
- ☐ A. Choosing items $\{1, 2\}$ for a benefit of 55
 - ☐ B. Choosing items $\{1, 3\}$ for a benefit of 58
 - ☐ C. Choosing items $\{1, 2, 3\}$ for a benefit of 72
 - ☒ D. Choosing items $\{1, 4\}$ for a benefit of 71
 - ☐ E. Choosing items $\{2, 3, 4\}$ for a benefit of 61

15. What is the optimal benefit in the following instance of the **Fractional Knapsack Problem** with four items where the maximum allowed weight is $W_{\max} = 10$. Below, i refers to item number, b_i denotes the benefit of item i and w_i denotes the weight of item i .

i	1	2	3	4
b_i	41	14	17	30
w_i	6	2	3	4

- ☐ A. 71
- ☐ B. 72
- ☐ C. 61
- ☐ D. 102
- ☒ E. $71 + \frac{1}{3}$